



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO COCOKATTUKOTTAI

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

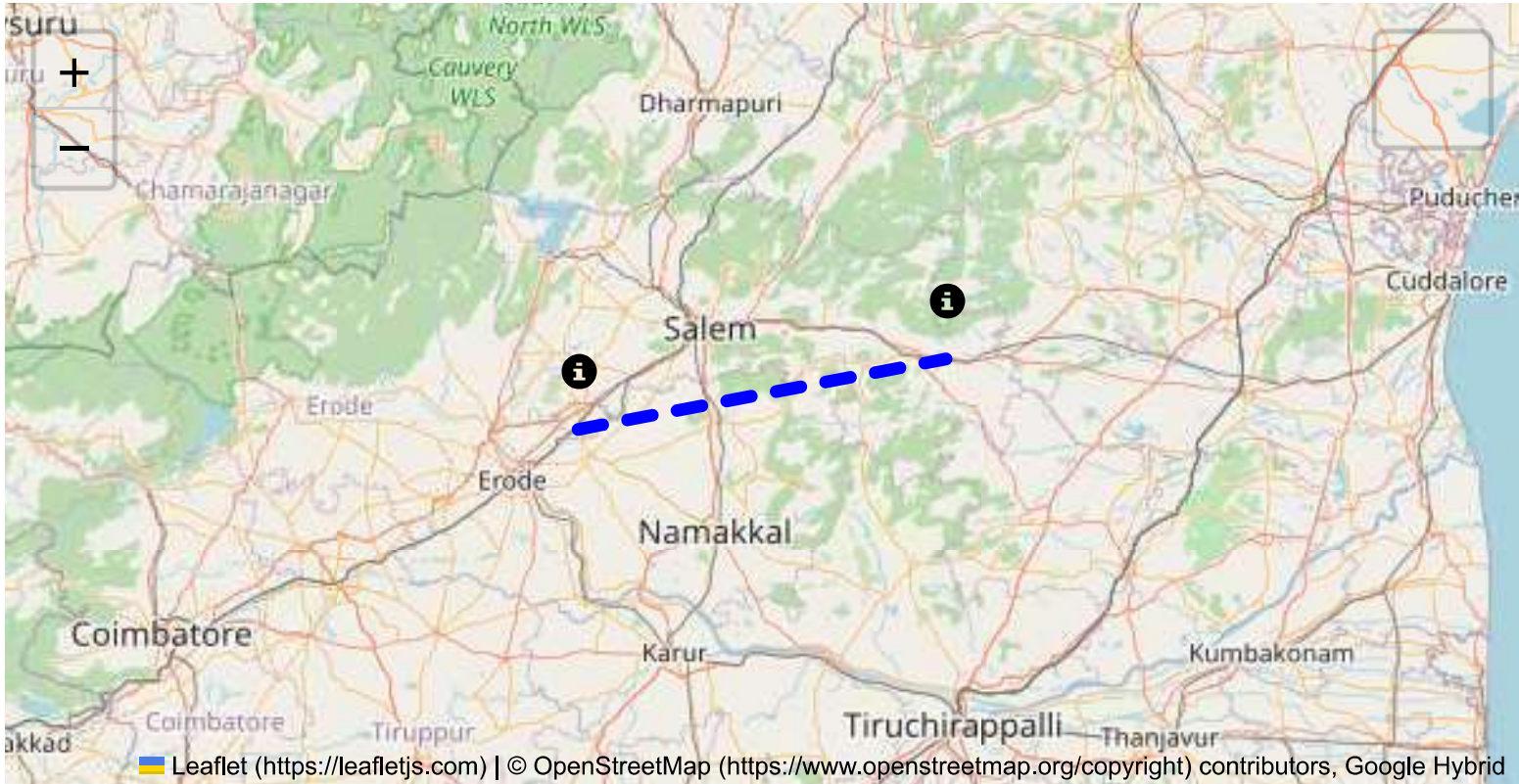
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

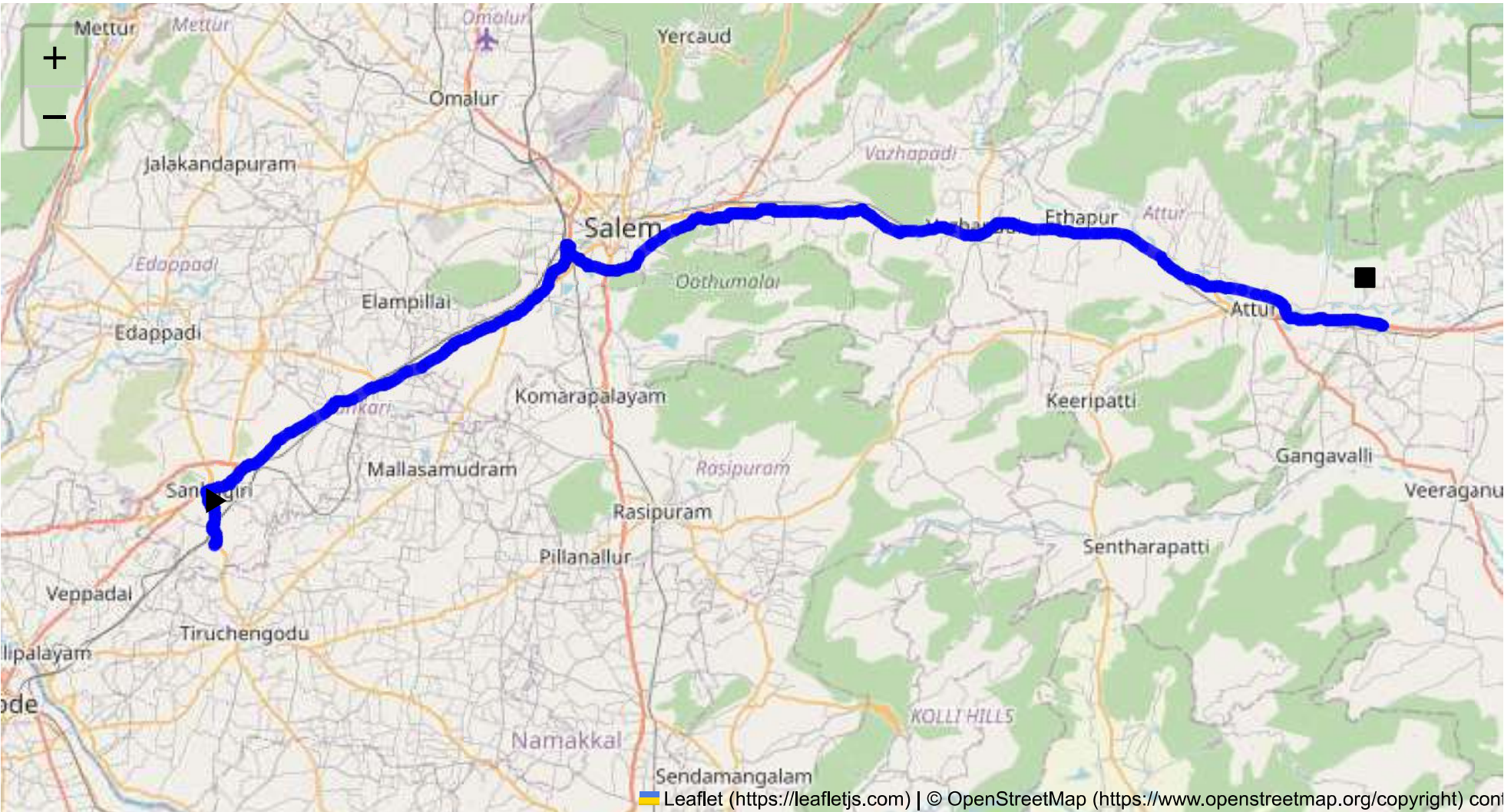
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 107.77 km
Estimated Duration: 2.0 hours
Adjusted Duration (Heavy Vehicle): 2.5 hours
Start: (11.4381, 77.8734)
End: (11.590355, 78.671697)



Welcome to the Journey Risk Management Study

To provide a comprehensive analysis of the route from CVQF+23W, Sangagiri, Tamil Nadu 637301, India, to HMRC+2MV, Vadachennimalai, Tamil Nadu 636121, India, here is a detailed breakdown based on each required point:

1. Overview of the Route Map:

- The route generally follows the NH-44 (previously known as NH-7), a major national highway that provides a direct connection between the start and end points. The route includes stretches through both urban areas and more rural settings.

2. Typical Weather Conditions and Potential Weather-Related Hazards:

- Tamil Nadu experiences a tropical climate with hot summers, mild winters, and significant rainfall during the monsoon season (October to December). Monsoon rains can lead to slippery roads, reduced visibility, and flash flooding, particularly in low-lying areas.

3. Traffic Patterns:

- The NH-44 is a busy highway with consistent traffic flow. Peak congestion typically occurs during morning (8:00 AM - 10:00 AM) and evening rush hours (5:00 PM - 8:00 PM). Traffic slows down near major towns such as Salem.

4. Assessment of Road Quality and Infrastructure:

- NH-44 is generally well-maintained, with paved surfaces and clear lane markings. However, occasional potholes and construction work may be encountered, especially after rains. Watch for uneven patches and poorly lit sections at night.

5. Suggestions for Alternative Routes for Emergencies:

- In case of major roadblocks on NH-44, consider using SH-86 for bypassing congested areas or accidents, though this may increase travel time.

6. Summary of Local Regulations Affecting Hazardous Material Transport:

- Transport of hazardous materials is regulated under the Motor Vehicles Act in India, requiring appropriate placarding and documentation. Transporters should ensure compliance with weight restrictions and have emergency management plans in place.

7. Overview of Historical Incidents:

- Historical incidents involving heavy vehicles often occur on bends or during adverse weather. Previous reports suggest avoiding overtaking on sharp curves and maintaining speed limits.

8. Environmental Considerations and Sensitive Areas:

- The route passes near several cultivated farmlands and water bodies. Spills or leaks could adversely affect the local environment, requiring immediate containment and cleanup measures.

9. Analysis of Communication Coverage:

- Mobile network coverage is generally strong along NH-44. However, some rural stretches may experience weak signals or dead zones, particularly in valley regions.

10. Estimated Emergency Response Times:

- Near urban centers like Salem, emergency response times typically range from 15 to 30 minutes. In more remote or rural segments, response times could extend to an hour or more, depending on severity and accessibility.

12. Overall Summary of Risk Assessment:

- While the route from Sangagiri to Vadachennimalai is generally safe for transporting hazardous materials, potential risks include adverse weather, road construction, and heavy traffic in urban areas. Regular updates on road conditions and weather forecasts, along with adherence to safety regulations, considerably mitigate these risks. It’s advisable to have contingency plans and communication devices ready for emergency situations.

Overall, careful planning and adherence to safety protocols will ensure a safe journey along this route for truck drivers carrying hazardous materials.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63303, 78.12713	15 KM/Hr
1	U-Turn	High	11.5882258, 78.6850455	10 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	Medium	11.44900, 77.87406	30 KM/Hr
5	Turn	Medium	11.45000, 77.87213	30 KM/Hr
6	Turn	High	11.45028, 77.87202	15 KM/Hr
7	Turn	Medium	11.45309, 77.87325	30 KM/Hr
8	Turn	Medium	11.46400, 77.87323	30 KM/Hr
9	Turn	Medium	11.46765, 77.87037	30 KM/Hr
10	Turn	High	11.47403, 77.86841	15 KM/Hr
11	Turn	High	11.49149, 77.89601	15 KM/Hr
12	Turn	High	11.49224, 77.89601	15 KM/Hr
13	Turn	Medium	11.64140, 78.11801	30 KM/Hr
14	Turn	Medium	11.64179, 78.11822	30 KM/Hr
15	Turn	Medium	11.64218, 78.11881	30 KM/Hr
16	Turn	Medium	11.64192, 78.12133	30 KM/Hr
17	Turn	Medium	11.64156, 78.12162	30 KM/Hr
18	Turn	Medium	11.64116, 78.12168	30 KM/Hr
19	Turn	Medium	11.63934, 78.12082	30 KM/Hr
20	Turn	Medium	11.65154, 78.35007	30 KM/Hr
21	Turn	Medium	11.59340, 78.62042	30 KM/Hr
22	Blind Spot	Blind Spot	11.58823, 78.68505	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
1	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
2	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
3	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
4	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
5	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
7	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
8	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
10	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
11	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
12	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
13	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
14	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
17	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
21	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium
23	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium
24	hospital	AVM Hospital	11.6264369, 78.1394204	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
25	hospital	மருத்துவமனை நிலம்	11.625344, 78.146058	30 km/h	Medium
26	hospital	Dharan Hospital	11.6252302, 78.1451753	30 km/h	Medium
27	hospital	Udhayam Hospital	11.625819, 78.145958	30 km/h	Medium
28	pharmacy	மருந்தகம்	11.6251501, 78.1470098	30 km/h	Medium
29	hospital	Rps Hospital	11.6260528, 78.1474921	30 km/h	Medium
30	hospital	Akitha Hospital	11.6256027, 78.1489875	30 km/h	Medium
31	hospital	Deepam Speciality Hospital	11.6269734, 78.1476357	30 km/h	Medium
32	hospital	Manima Clinic	11.623248, 78.1465721	30 km/h	Medium
38	hospital	Jaaidev Hospital	11.6655205, 78.2634736	30 km/h	Medium
40	clinic	Lavanya Clinic	11.652121, 78.4795146	30 km/h	Medium
41	hospital	Amsh Agathiyar Sidha Hospital	11.650564, 78.483542	30 km/h	Medium
42	clinic	Sudha Polyclinic	11.647825, 78.508537	30 km/h	Medium
43	hospital	Government Hospital, Kothampadi	11.62641, 78.542199	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
6	marketplace	Sunday Market	11.4796502, 77.8708458	30 km/h	Medium
15	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
16	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium